

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
FINAL YEAR (COMPUTER SCIENCE/
COMPUTER SCIENCE SPECIALIZATION IN ARTIFICIAL INTELLIGENCE)
FALL SEMESTER EXAMINATIONS 2025
BATCH 2022

Time: 3 Hours

Dated: 04-DEC-25
Max.Marks:60

Natural Language Processing – CT-485

Note: Attempt all questions.

Ms. DureShahwar

Q1 (a) Discuss the elements of semantic analysis: Hyponymy, Homonymy, Polysemy, Synonymy, and Antonymy with examples.
(b) Explain Semantic Frames and Semantic Roles with examples.

[CLO-1: 12 Marks]

Q2 (a) Discuss the problem of Zero Frequency in Naive Bayes and explain how Laplace Smoothing solves it.
(b) Compare the sigmoid, tanh, and ReLU activation functions in terms of their output behavior.

[CLO-1: 12 Marks]

Q3. Explain how biased language models can reinforce stereotypes, cause discrimination, propagate misinformation and affect inclusivity. Illustrate your answer with examples from hiring systems, criminal justice, or finance.

[CLO-1: 6 Marks]

Q4. Analyze the following example case related to the Bag-of-Words model.

- *NLP is a powerful tool*
- *Python makes NLP easier*

- a) Create a Bag of Words (BoW) representation for these two sentences.
- b) Calculate the TF (Term Frequency) for each word.
- c) Calculate the IDF (Inverse Document Frequency) for each word.
- d) Compute the TF-IDF values for each word.

[CLO-2: 6 Marks]

Q5. (a) Analyze the following corpus and demonstrate the difference between Stemming and lemmatization outputs.

"The athletes are running, ran, and will run in the championship next week. Meanwhile, coaches are training, trained, and have trained their teams for various competitions throughout the year."

- (b) Explain why the results differ and how each technique treats the words.

[CLO-2: 6 Marks]

Q6. Apply the ROUGE-2 (Recall-Oriented Understudy for Gisting Evaluation) technique to assess the of quality a system-generated summary by comparing it with human reference summaries.

[CLO-3: 6 Marks]

Candidate (System) Summary:

Reinforcement learning enables robots to learn tasks autonomously.

Reference (Human) Summaries:

Human 1: Robots can learn tasks autonomously through reinforcement learning.

Human 2: Reinforcement learning allows robots to acquire skills without supervision.

Q7. Apply a bigram language model to the following corpus and answer the Questions given below.

[CLO-3: 6 Marks]

Training data:

- <s> machine learning improves </s>
- <s> deep learning models </s>
- <s> machine models learn </s>

1. Predict the most probable next word for:

- (1) <s> machine ...
- (2) <s> deep learning ...

2. Which of the following sentences is better, i.e., gets a higher probability with this model?

- (1) <s> machine learning models </s>
- (2) <s> deep learning improves </s>

Q.8. Choose the most appropriate evaluation metric (accuracy, precision, recall, or F1-score) for evaluating a spam-filter system where the dataset is highly imbalanced (only 5% of emails are spam, 95% are non-spam). Justify your choice clearly.

[CLO-3: 6 Marks]

Time: 3 Hours

Dated: 23-01-2024
Max.Marks: 60

Natural Language Processing - CT-485

Instructions:

Attempt all questions

All questions carry equal marks

QUESTION#1

(CLO-3,12 marks)

DEMONSTRATE suitable methods of aspect-based sentiment analysis for the given customer reviews;

- "The food at this restaurant is absolutely divine! Every dish is crafted with precision, bursting with flavors that take my taste buds on a delightful journey. The chef's dedication to quality ingredients is truly commendable."
- "I had high expectations for the service, but it fell short. The staff seemed disorganized, and our orders took forever to arrive. Disappointing experience considering the restaurant's reputation."
- "While it's not the cheapest place in town, the value for money is evident in the generous portions and high-quality ingredients. The prices are justified for the exquisite dining experience."
- "I was quite put off by the lack of cleanliness in the dining area. Dirty tables and unattended spills made the overall dining experience less enjoyable. Hygiene should be a top priority."
- "While it's not the cheapest place in town, the value for money is evident in the generous portions and high-quality ingredients. The prices are justified for the exquisite dining experience."
- "The menu here is a gastronomic adventure! From traditional dishes to innovative creations, there's something for every palate. The variety ensures that each visit is a unique culinary journey."
- "The restaurant's reservation system needs improvement. Despite having a reservation, we had to wait in line, and the staff seemed unaware of our booking. A frustrating experience."

QUESTION#2

(CLO-2,12 marks)

EXAMINE the given dataset to assess similarity between words using cosine similarity measurements. Also, **ANALYSE** the similarity measures for every possible combination.

	data	classes	noise	preprocessing	rice
information	348	2011	89	119	6
food	14	1	9	10	804
Knowledge	561	1100	654	1683	15
text	507	3352	223	2670	5
learning	402	752	999	1023	5

P.T.O

QUESTION#3

(CLO-1,12 marks)

Annotating a gold standard corpus is a crucial step in natural language processing tasks. **DESCRIBE** the key considerations and challenges involved in annotating a gold standard corpus (Consider dataset of Question-1). Also, **DISCUSS** the criteria of selecting the annotators and evaluation of gold standard corpus.

QUESTION#4

(CLO-3,12 marks)

"The domain of semantics is subject to ambiguity". **DEMONSTRATE** this expression (Use the given dataset for examples). **USE** suitable methods for identification of 'Named Entity Recognition' application.

Headline: Technology Giants Unveil Collaborative Project for Sustainable Energy Solutions
In a groundbreaking announcement today, leading technology companies Apple, Google, and Microsoft revealed their joint initiative aimed at developing innovative solutions for sustainable energy. The collaboration, named "GreenTech Innovators," will focus on harnessing renewable resources to power data centers and reduce the carbon footprint of the tech industry. The CEOs of Apple, Tim Cook, Google's Sundar Pichai, and Microsoft's Satya Nadella, expressed their commitment to driving positive environmental change through this unprecedented partnership.

Headline: Global Health Organizations Unite to Combat Emerging Infectious Diseases
Amid growing concerns over the threat of emerging infectious diseases, major global health organizations, including the World Health Organization (WHO), the Centers for Disease Control and Prevention (CDC), and Médecins Sans Frontières (MSF), have formed a united front. This collaborative effort, named "GlobalHealth Unity," aims to enhance international cooperation, share critical data, and deploy rapid response teams to mitigate the impact of pandemics. Dr. Tedros Adhanom Ghebreyesus, Director-General of WHO, emphasized the importance of collective action in safeguarding global health.

Headline: Political Leaders Gather for Historic Peace Summit in the Middle East
In a historic development, political leaders from across the Middle East convened for a peace summit to address long-standing conflicts in the region. The summit, facilitated by the United Nations, witnessed the participation of prominent figures, including Israeli Prime Minister Benjamin Netanyahu, Palestinian President Mahmoud Abbas, and Saudi Arabian Crown Prince Mohammed bin Salman. The leaders engaged in constructive dialogue, expressing a shared commitment to fostering stability and cooperation in the region.

QUESTION#5

(CLO-2,12 marks)

QUANTIFY signature words using Term Frequency - Inverse Document Frequency (TF-IDF) technique in the given dataset. Also, **ANALYZE** the output matrix.

Document 1: Artificial Intelligence is reshaping industries.

Document 2: Companies worldwide are investing in Artificial Intelligence technologies to gain a competitive edge.

Document 3: The stock market experienced a downturn today due to some companies.

Document 4: Artificial Intelligence in medical research has led to breakthroughs in disease treatment.

Time: 3 Hours

Dated: 03-02-2023
Max. Marks: 60

Natural Language Processing - CT-485

Instructions:

- Attempt all questions
- All questions carry equal marks
- Use assumptions where required

[CLO-1, 12 Marks]

Question#1
Consider an example of a large dataset (sample dataset is provided) that contains reflections of students on specific assignment. Teacher wants to assign grades automatically based on certain rubrics (given). Due to large number of reflections, teacher is facing difficulty to assess manually and wants an automatic method to assign grades to these reflections. WHICH information should be extracted for automatic grading system) and WHAT will be your approach to assign grades based on rubrics?

Q. Why dictionaries cannot be sufficient for information extraction if all the variants of words will be stored?	Rubric: Explanation with specific examples (+2 marks) General explanation without example (+1 mark)
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SAMPLE DATASET ANSWERS FROM STUDENTS

CT-001 We can't store combination of roots, stems, morpheme in our dictionary because storing these combination would be very costly in terms of storage capacity as well as in retrieval. For example: If we develop a chat bot which uses this dictionary containing these combination, so how much CPU intensive it would be to handle a real user in quickly answering his queries.
CT-008 Reason: because it will be very time consuming and will also take up a lot of space. The major reason can be that using a dictionary for every possible use case will result in similar outputs that can occur as a result of repetition of same words in 2 or more dictionaries, hence leading to inefficient results.
CT-048 Nope, we can't! 1- From the technical perspective: Since there are thousands of root words are there and a combination of these words or variants (all verb forms, plural form, gerund, morphological rules etc) would result in a humongous amount which cannot be retrieved in a single block of memory. 2- Ambiguity of words meanings can't be just handle in storing them in a dictionary, for instance, "Apple" could be a fruit or a company.
CT-11 Storing every possible combinations which are endless would require huge amount of space and it will be difficult for the system to retrieve it hence making the system totally inefficient and sluggish. If it is implemented then the hardware resources will be used to a great extent and will yield inefficient results.
CT-005 Technically if we store every possible words. Then there will be the points which we need to cater and can become a problem. 1) Large amount of memory 2) Duplication of words 3) Low processing speed as data will be large. 4) It will require more hardware storage 5) Less efficient due to low retrieval of data.
CT-038 No we cannot because we need large amount of space to store them and required huge computational resources to parse by matching all combinations. And we certainly cannot list all the morphological variants of every words of complex languages like Turkish, Chinese. We can solve the parsing issue by using finite state transducer. FST takes input words and outputs their stems and modifiers.

[CLO-3, 12 Marks]

Question#2

- a) "Careful feature selection may affect the performance of text classification task". Considering that may be effective for classification of

CT-005 Technically if we store every possible words. Then there will be the points which we need to cater and can become a problem. 1) Large amount of memory 2) Duplication of words 3) Low processing speed as data will be large. 4) It will require more hardware storage 5) Less efficient due to low retrieval of data.

CT-038 No we cannot because we need large amount of space to store them and required huge computational resources to parse by matching all combinations. And we certainly cannot list all the morphological variants of every words of complex languages like Turkish, Chinese. We can solve the parsing issue by using finite state transducer. FST takes input words and outputs their stems and modifiers.

Question#2

[CLO-3, 12 Marks]

- a) "Careful feature selection may affect the performance of text classification task". Considering this statement, **IDENTIFY** some of the features that may be effective for classification of aspects in reviews (related to restaurants) posted on social media. (4)

- b) **APPLY** Naïve Bayes Classification on the following data set. (8)

Training	Document #	Words	Class
	1	Polymorphism Inheritance Object	OOP
	2	Semantics Vector Analysis	NLP
	3	Object Classes Inheritance Language	OOP
	4	Programming Language Polymorphism	OOP
	5	Semantics Classification Analysis Object	NLP
Test	6	Vector Semantics Analysis Classification	?
	7	Programming Classes Inheritance Vector	?

P.T

Question#3

[CLO-2, 12 Marks]

ILLUSTRATE the application pipeline of "Aspect Based Sentiment Analysis-ABSA" for the given dataset.

As a diehard Samsung fan who has had every Samsung phone since before the S series started, this has been my favorite upgrade so far. The battery life is amazing.
Adequate feel. Nice heft. Processor's still sluggish for the few apps that I have installed. Samsung wants to vacuum all of your data with their apps that requires more access than needed. What else is new.
Never disappointed. One of the reasons I've been a long Samsung (product) consumer, they're ability to impress me with each product release. I've owned, or maintained, all Samsung smartphones starting with the S4 and the S8 is the best one yet.
I've now found that I'm in a group of people that have carried their phone in their pocket without problems until the S8. Day one screen has crack from being in my pocket. Good news.....battery life is great and lots of great features.
I am the type of person who never would completely be sold on a phone I really was not into Samsung until the S7 (Before that I had a Galaxy Nexus, but that was a Google phone). I got the S7 because at the time the fries made the most sense for me.
The way this Samsung S8 phone operates is more cumbersome than my previous phone, HTC M9. When making a call you must select phone and then select the dial pad. It should automatically go to the dial pad.
I bought this phone-very excited to use it. I agree with all the other reviews stating that trying to use a glass screen protector with it is a nightmare. I have so far has 2 different screen protectors on it and both times they have made it difficult for me to tap on the screen.
It is an extremely advanced and truly a Smart phone. Great apps, great look. BUT, the durability is that of a fresh egg. The screen is fragile and one crack voids warranty. Won't buy another Samsung.
Great phone with a PHENOMENAL Camera, not all that hard to get use to. However the screen IS NOT 5.8," it's not all that bigger than my S7 Edge. So maybe a 5.6" screen, it just looks bigger, takes amazing pictures.
I initially wasn't 100% certain about this handset, coming from the Nexus line I am very much used to a stock Android OS, I was worried about bloatware and various other things Samsung would add.
I do not like this phone. I had it in my pocket with something else and the screen protector cracked. Replaced it and it has another crack in it. They need to do something different.

Question#4

[CLO-2, 12 Marks]

SHOW the complete working of Term Frequency— Inverse Document Frequency (TF-IDF) for the given dataset and SUMMARISE your results.

Document 1: On Wednesday, Kate ate an orange. $\Rightarrow 76$
 Document 2: I like orange. $\Rightarrow$

It's not all that different from my screen, it just looks bigger, takes amazing pictures. I initially wasn't 100% certain about this handset, coming from the Nexus line I am very much used to a stock Android OS. I was worried about bloatware and various other things Samsung would add. I do not like this phone. I had it in my pocket with something else and the screen protector cracked. Replaced it and it has another crack in it. They need to do something different.

Question#4

[CLO-2, 12 Marks]

SHOW the complete working of Term Frequency— Inverse Document Frequency (TF-IDF) for the given dataset and SUMMARISE your results.

Document 1: On Wednesday, I ate an orange. $\Rightarrow 6$
 Document 2: I like orange. $\Rightarrow$
 Document 3: I ate apple and orange on Wednesday. $\Rightarrow$
 Document 4: Day before Wednesday, I ate an apple. $\Rightarrow$

Question#5

[CLO-3, 12 Marks]

a) For the given measurements, UTILISE Inter-Annotator Agreement (IAA) for the evaluation of gold standard dataset. (4)

	Matches	Non-Matches
Annotator 1	3000	Your Roll No. + 15
Annotator 2	2899	Your Roll No. + 20
Annotator 3	2560	Your Roll No. + 30

b) FIND Cosine similarity of the given word vectors to relate word similarities. (8)

	Context 1	Context 2	Context 3	Context 4	Context 5
Word 1	15	78	0	1	265
Word 2	82	15	14	0	9
Word 3	230	25	4	4	19

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NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY
BSCT/BSCT(AI), Midterm Examination Fall 2025

Time: 90 minutes

Natural Language Processing (CT-485) – A

Max Marks: 20

Note: Attempt all questions.

- Q1) **[CLO-2]** Discuss techniques for Sentiment Analysis, Part of Speech Tagging (POS), Text Classification, and explain their importance in Natural Language Processing. [5 Marks]
- Q2) **[CLO-3]** Apply top-down parsing to the sentences below, and write the step-by-step derivation. [5 Marks]
1. "NLP predicts sentiment".
Grammar Rule: $S \rightarrow NP VP, NP \rightarrow N, VP \rightarrow V NP, N \rightarrow \text{"NLP"} \mid \text{"sentiment"}, V \rightarrow \text{"predicts"}$
 2. "The system detects anomalies".
Grammar Rule: $S \rightarrow NP VP, NP \rightarrow Det N, VP \rightarrow V NP, Det \rightarrow \text{"The"}, N \rightarrow \text{"system"} \mid \text{"anomalies"}, V \rightarrow \text{"detects"}$
- Q3) **[CLO-3]** Apply appropriate NER technique on the given sentences with BIO Tagging. [5 Marks]
1. "Elon Musk announced a new project in San Francisco on Friday".
 2. "Professor Alan Turing presented a lecture on Artificial Intelligence at Cambridge University in 1950".
- Q4) **[CLO-4]** Analyze below the raw corpus sample: [5 Marks]
- "um I think NLP is... llke very cool!!! but... u know it needs CLEANING??"
1. Identify at least four preprocessing challenges in this text.
 2. Explain why addressing these challenges is important for building reliable NLP models.

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NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY
BSCT/BSCT(AI), Midterm Examination Fall 2025

Time: 90 minutes

Natural Language Processing (CT-485) – B

Max Marks: 20

Note: Attempt all questions.

Q1) [CLO-2] Discuss techniques for Text Summarization, Machine Translation, Named Entity Recognition (NER), and explain their importance in Natural Language Processing. [5 Ma

Q2) [CLO-3] Apply top-down parsing to the sentences below, and write the step-by-step derivation. [5 Ma

1. "Chabot answer questions".

Grammar Rule: $S \rightarrow NP VP$, $NP \rightarrow N$, $VP \rightarrow V NP$, $N \rightarrow \text{"Chatbot"} \mid \text{"questions"}$, $V \rightarrow \text{"answer"}$

2. "The algorithm learns patterns".

Grammar Rule: $S \rightarrow NP VP$, $NP \rightarrow Det N$, $VP \rightarrow V NP$, $Det \rightarrow \text{"The"}$, $N \rightarrow \text{"algorithm"} \mid \text{"patterns"}$, $V \rightarrow \text{"learns"}$

Q3) [CLO-3] Apply appropriate NER technique on the given sentences with BIO Tagging. [5 M

1. "Bill Gates spoke at the World Health Conference in Geneva in 2018".

2. "NASA launched the Artemis mission from Kennedy Space Center on November 16, 2022".

Q4) [CLO-4] Analyze below the raw corpus sample: [5 M

"RT@user123: The climate-change!!! report... was released today?? link www.news.com"

1. Identify at least four preprocessing challenges in this text.

2. Explain why addressing these challenges is important for building reliable NLP models.

C1-22

Paper-A

Natural Language Processing (CT-485)

Midterm Date: 28-11-2023

Instructions:

Duration: 90 minutes

Attempt all questions.

Make necessary assumptions if required.

Q1. EXPLAIN step by step system flow of named entity recognition system for the given dataset with diagram. Also, DISCUSS the possibility of ambiguity in the given dataset. (CLO-2, 12 marks)

Apple Inc. is planning to open a new store in Paris.

Jane Smith, a renowned scientist, won the Nobel Prize in Physics.

The river Nile is the longest river in Africa.

Google is developing a new artificial intelligence technology.

On January 20, 2021, Joe Biden was inaugurated as the President of the United States.

The Mona Lisa, painted by Leonardo da Vinci, is displayed at the Louvre Museum.

Microsoft Corporation was founded by Bill Gates and Paul Allen.

The Eiffel Tower is a famous landmark located in Paris, France.

In 1969, Neil Armstrong became the first person to walk on the moon.

The Great Wall of China is one of the wonders of the world.

Q.2 IMPLEMENT Naïve Bayes algorithm on the given dataset for text classification task.

(CLO-3, 8 marks)

	Document #	Words	Class
Training	1	SVM, Clustering Supervised Unsupervised	DM
	2	Requirements Analysis Prototype Agile, Design	SAD
	3	Prototype <u>NaïveBayes</u> Regression Prediction Unsupervised	DM
	4	SVM <u>NaïveBayes</u> Regression Classification Supervised	DM
	5	Agile Scrum Analysis Design	SAD
Test	6	NaïveBayes Regression Classification Supervised	?
	7	Analysis Classification Agile	?

$|V| =$
 $DM = 14$
 $|V| = 14$
 $SAD = 9$

SVM, Clustering, Supervised, Unsupervised,
 Req, Ana, Proto, Agile, Des, NvB, Reg, Class,
 Scrum, Prediction

Instructions:

Duration: 90 minutes

Attempt all questions.

Make necessary assumptions if required.

Q.1 EXPLAIN step by step process flow of named entity recognition system for the given dataset with diagram.

Also, DISCUSS the possibility of ambiguity in the given dataset.

(CLO-2, 12 marks)

A recent study published in the Journal of Medicine suggests that aspirin can reduce the risk of heart attacks.

Dr. Emily Johnson, a leading oncologist, recommends regular screenings for breast cancer.

The patient was prescribed 20mg of Lipitor to manage cholesterol levels in pt.

The CDC advises getting vaccinated against Influenza every flu season.

In a groundbreaking discovery, researchers identified a new gene linked to Alzheimer's disease.

The surgery, performed by Dr. Michael Rodriguez, was a success.

The FDA approved a new drug for the treatment of diabetes.

Patients (pt) with hypertension should monitor their blood pressure regularly.

The World Health Organization (WHO) recommends a balanced diet for overall well-being.

A recent case study highlighted the effectiveness of immunotherapy in cancer treatment.

Q.2 IMPLEMENT Naïve Bayes algorithm on the given dataset for text classification task.

(CLO-3, 8 marks)

	Doc #	Words	Class
Training	1	Story was disappointing and depressing	Negative
	2	Hero played bad	Negative
	3	Waste of time	Negative
	4	Liked acting	Positive
	5	Loved this story superb characters	Positive
Test	6	Liked story and characters	?
	7	Complete time waste	?

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
Mid-Term Examination (DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION
TECHNOLOGY)

Fall Semester Examinations 2020

Dated: 06-12-2022

Max. Marks: 20

Duration: 90 minutes

Instructions:

- Attempt all questions.
- Each question carry equal marks.
- Use A4 sheet to solve your paper, write Roll# and page# on every page.

(CLO-2, 10 marks)

Question#1

Consider the following data; CLASSIFY part-of-speech tags using Hidden Markov Model (show complete steps) to test the following sequences; 1. May will see John. 2. Will will play with Peter. DEMONSTRATE your understanding towards the limitations of hand-coded patterns.	I P see V John N in Prp May N Peter N will V play V with V Prp May N May N and CC Will N are V friends N John N may V see V May N Where, N: Noun, P: Pronoun, Prp: Preposition, CC: Conjunction
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Question#2

(CLO-3, 10 marks)

WHAT is the importance of "Annotation Process" in Natural Language Processing? Consider the following data set and DEVELOP comprehensive annotation guidelines (using a manual or semi-automatic approach) for a novice user. Guidelines must be developed for the categories: Person, Location, Organization, Infrastructure, and Date. Your annotation guidelines must be clear to avoid ambiguity problems for all the labels/categories. Underline your information and mark it with the label using different colours or shapes (if you don't have colors). Write down some challenges you faced in the development of the annotation guidelines for the given dataset.

"Mrs. May visited Leeds General Infirmary hospital which is located near Burley Road. She was not feeling well and had appointment with Dr. Ray Johnson. Mrs. May was last seen by Dr. Parker in May 2019, who advised her Chest x-Ray. Dr. Ray examined her and by looking at her x-ray report, he advised her to quit smoking. She told him that she already worked very hard to quit it and use to have coffee instead every day from Burley Cafe. Dr. Johnson requested her to go to Radiography lab to get a new X-Ray so he can analyse her current situation. He also prescribed her medicines and ask her to collect it from Lloyds Pharmacy in front of Parkinson building and visit again on 13th June."